Solution Deployment and Validation Details

NetApp Solutions

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Table of Contents

Solution Deployment and Validation Details . . . 1
ONTAP AI Deployment . . . 1
Kubernetes Deployment . . . 1
cnvrg.io Deployment . . . 1

Solution Deployment and Validation Details

The following sections discuss the details of solution deployment and validation.

ONTAP AI Deployment

Deployment of ONTAP AI requires the installation and configuration of networking, compute, and storage hardware. Specific instructions for deployment of the ONTAP AI infrastructure are beyond the scope of this document. For detailed deployment information, see NVA-1121-DEPLOY: NetApp ONTAP AI, Powered by NVIDIA.

For this solution validation, a single volume was created and mounted to the DGX-1 system. That mount point was then mounted to the containers to make data accessible for training. For large-scale deployments, NetApp Trident automates the creation and mounting of volumes to eliminate administrative overhead and enable end-user management of resources.

Kubernetes Deployment

To deploy and configure your Kubernetes cluster with NVIDIA DeepOps, perform the following tasks from a deployment jump host:

1. Download NVIDIA DeepOps by following the instructions on the Getting Started page on the NVIDIA DeepOps GitHub site.
2. Deploy Kubernetes in your cluster by following the instructions on the Kubernetes Deployment Guide on the NVIDIA DeepOps GitHub site.

> For the DeepOps Kubernetes deployment to work, the same user must exist on all Kubernetes master and worker nodes.

If the deployment fails, change the value of `kubectl_localhost` to false in `deepops/config/group_vars/k8s-cluster.yml` and repeat step 2. The `Copy kubectl binary to ansible host` task, which executes only when the value of `kubectl_localhost` is true, relies on the fetch Ansible module, which has known memory usage issues. These memory usage issues can sometimes cause the task to fail. If the task fails because of a memory issue, then the remainder of the deployment operation does not complete successfully.

If the deployment completes successfully after you have changed the value of `kubectl_localhost` to `false`, then you must manually copy the `kubectl binary` from a Kubernetes master node to the deployment jump host. You can find the location of the `kubectl binary` on a specific master node by running the `which kubectl` command directly on that node.

cnvrg.io Deployment

Deploy cnvrg CORE Using Helm

Helm is the easiest way to quickly deploy cnvrg using any cluster, on-premises, Minikube, or on any cloud cluster (such as AKS, EKS, and GKE). This section describes how cnvrg was installed on an on-premises (DGX-1) instance with Kubernetes installed.

Prerequisites

Before you can complete the installation, you must install and prepare the following dependencies on your local machine:

- Kubectl
- Helm 3.x
- Kubernetes cluster 1.15+

Deploy Using Helm

1. To download the most updated cnvrg helm charts, run the following command:

```
helm repo add cnvrg https://helm.cnvrg.io
helm repo update
```

2. Before you deploy cnvrg, you need the external IP address of the cluster and the name of the node on which you will deploy cnvrg. To deploy cnvrg on an on-premises Kubernetes cluster, run the following command:

```
helm install cnvrg cnvrg/cnvrg --timeout 1500s  --wait \ --set
global.external_ip=<ip_of_cluster> \ --set global.node=<name_of_node>
```

3. Run the `helm install` command. All the services and systems automatically install on your cluster. The process can take up to 15 minutes.
4. The `helm install` command can take up to 10 minutes. When the deployment completes, go to the URL of your newly deployed cnvrg or add the new cluster as a resource inside your organization. The `helm` command informs you of the correct URL.

```
Thank you for installing cnvrg.io!
Your installation of cnvrg.io is now available, and can be reached via:
Talk to our team via email at
```

5. When the status of all the containers is running or complete, cnvrg has been successfully deployed. It should look similar to the following example output:

```
NAME                                READY   STATUS      RESTARTS   AGE
cnvrg-app-69fbb9df98-6xrgf              1/1     Running     0          2m
cnvrg-sidekiq-b9d54d889-5x4fc           1/1     Running     0          2m
controller-65895b47d4-s96v6             1/1     Running     0          2m
init-app-vs-config-wv9c4                0/1     Completed   0          9m
init-gateway-vs-config-2zbpp            0/1     Completed   0          9m
init-minio-vs-config-cd2rg              0/1     Completed   0          9m
minio-0                                 1/1     Running     0          2m
postgres-0                              1/1     Running     0          2m
redis-695c49c986-kcbt9                  1/1     Running     0          2m
seeder-wh655                            0/1     Completed   0          2m
speaker-5sghr                           1/1     Running     0          2m
```

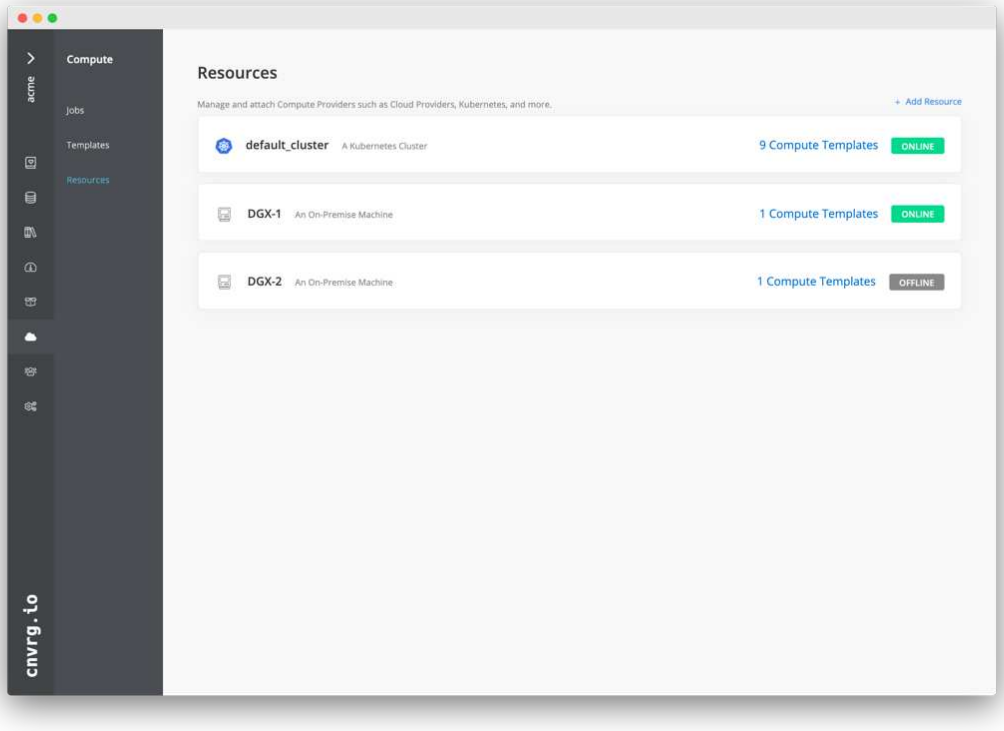
Computer Vision Model Training with ResNet50 and the Chest X-ray Dataset

cnvrg.io AI OS was deployed on a Kubernetes setup on a NetApp ONTAP AI architecture powered by the NVIDIA DGX system. For validation, we used the NIH Chest X-ray dataset consisting of de-identified images of chest x-rays. The images were in the PNG format. The data was provided by the NIH Clinical Center and is available through the NIH download site. We used a 250GB sample of the data with 627, 615 images across 15 classes.

The dataset was uploaded to the cnvrg platform and was cached on an NFS export from the NetApp AFF A800 storage system.

Set up the Compute Resources

The cnvrg architecture and meta-scheduling capability allow engineers and IT professionals to attach different compute resources to a single platform. In our setup, we used the same cluster cnvrg that was deployed for running the deep-learning workloads. If you need to attach additional clusters, use the GUI, as shown in the following screenshot.



Load Data

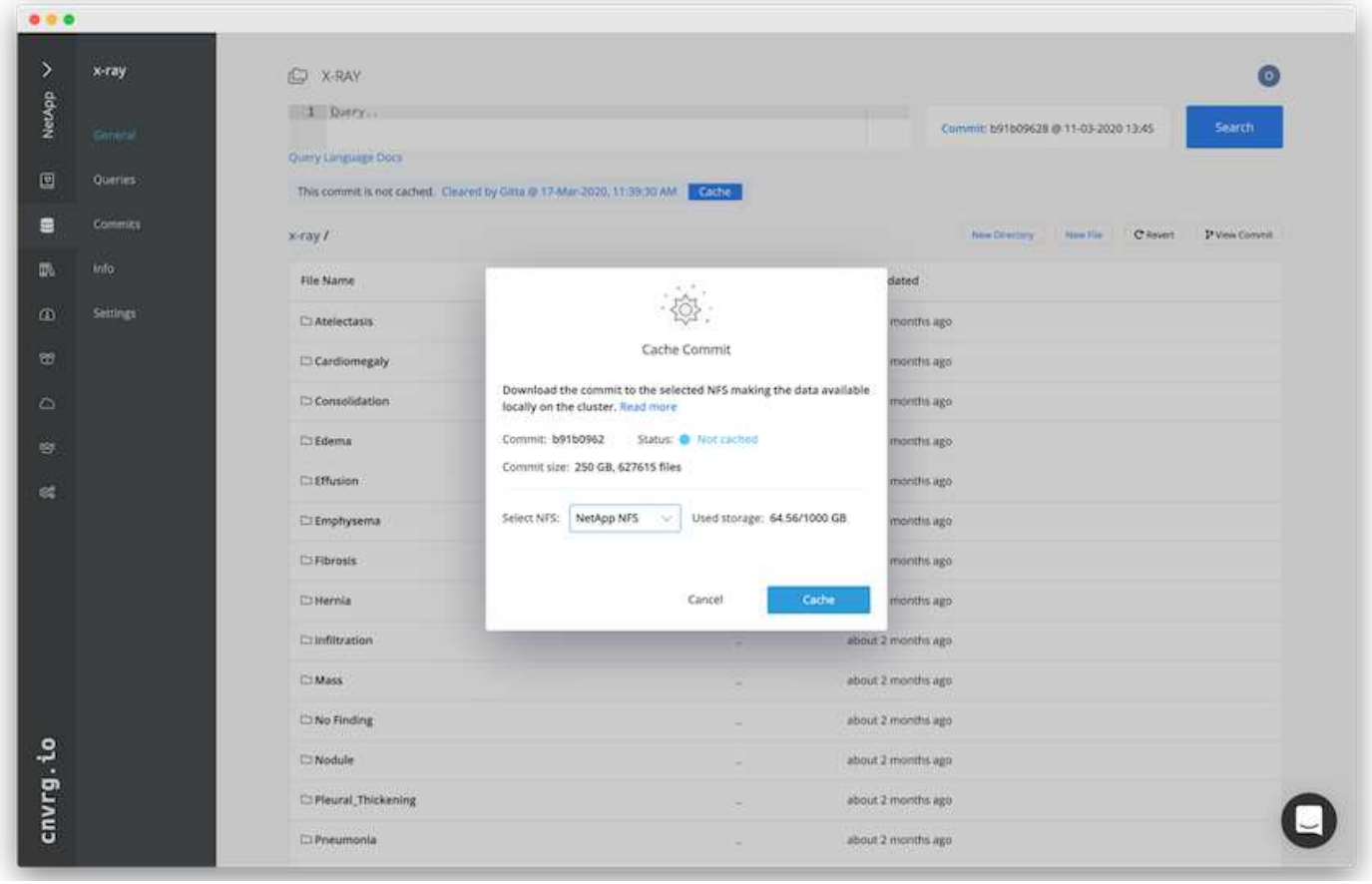
To upload data to the cnvrg platform, you can use the GUI or the cnvrg CLI. For large datasets, NetApp recommends using the CLI because it is a strong, scalable, and reliable tool that can handle a large number of files.

To upload data, complete the following steps:

1. Download the cnvrg CLI.
2. navigate to the x-ray directory.
3. Initialize the dataset in the platform with the `cnvrg data init` command.
4. Upload all contents of the directory to the central data lake with the `cnvrg data sync` command.After the data is uploaded to the central object store (StorageGRID, S3, or others), you can browse with the GUI. The following figure shows a loaded chest X-ray fibrosis image PNG file. In addition, cnvrg versions the data so that any model you build can be reproduced down to the data version.

Cach Data

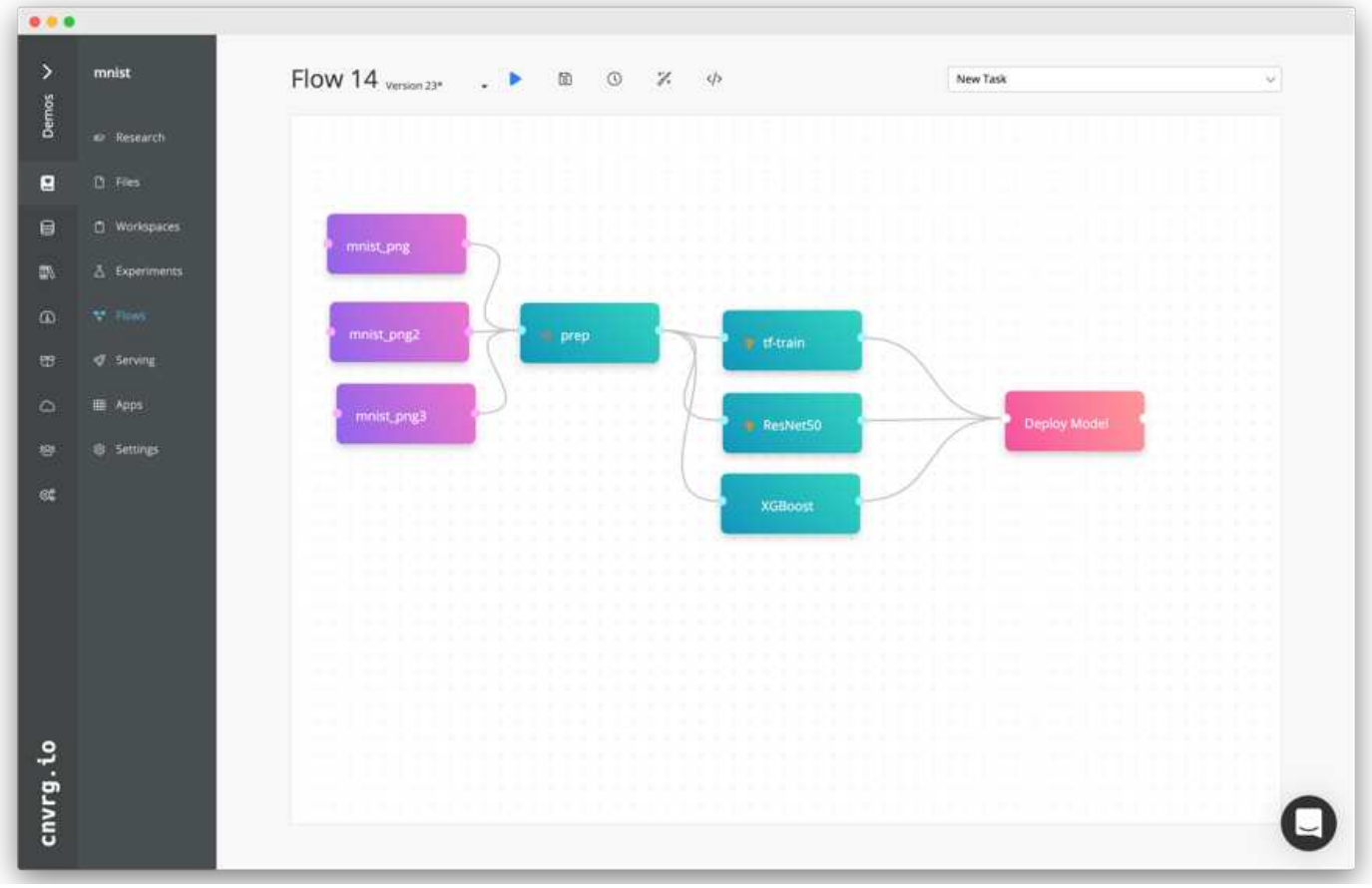
To make training faster and avoid downloading 600k+ files for each model training and experiment, we used the data-caching feature after data was initially uploaded to the central data-lake object store.



After users click Cache, cnvrg downloads the data in its specific commit from the remote object store and caches it on the ONTAP NFS volume. After it completes, the data is available for instant training. In addition, if the data is not used for a few days (for model training or exploration, for example), cnvrg automatically clears the cache.

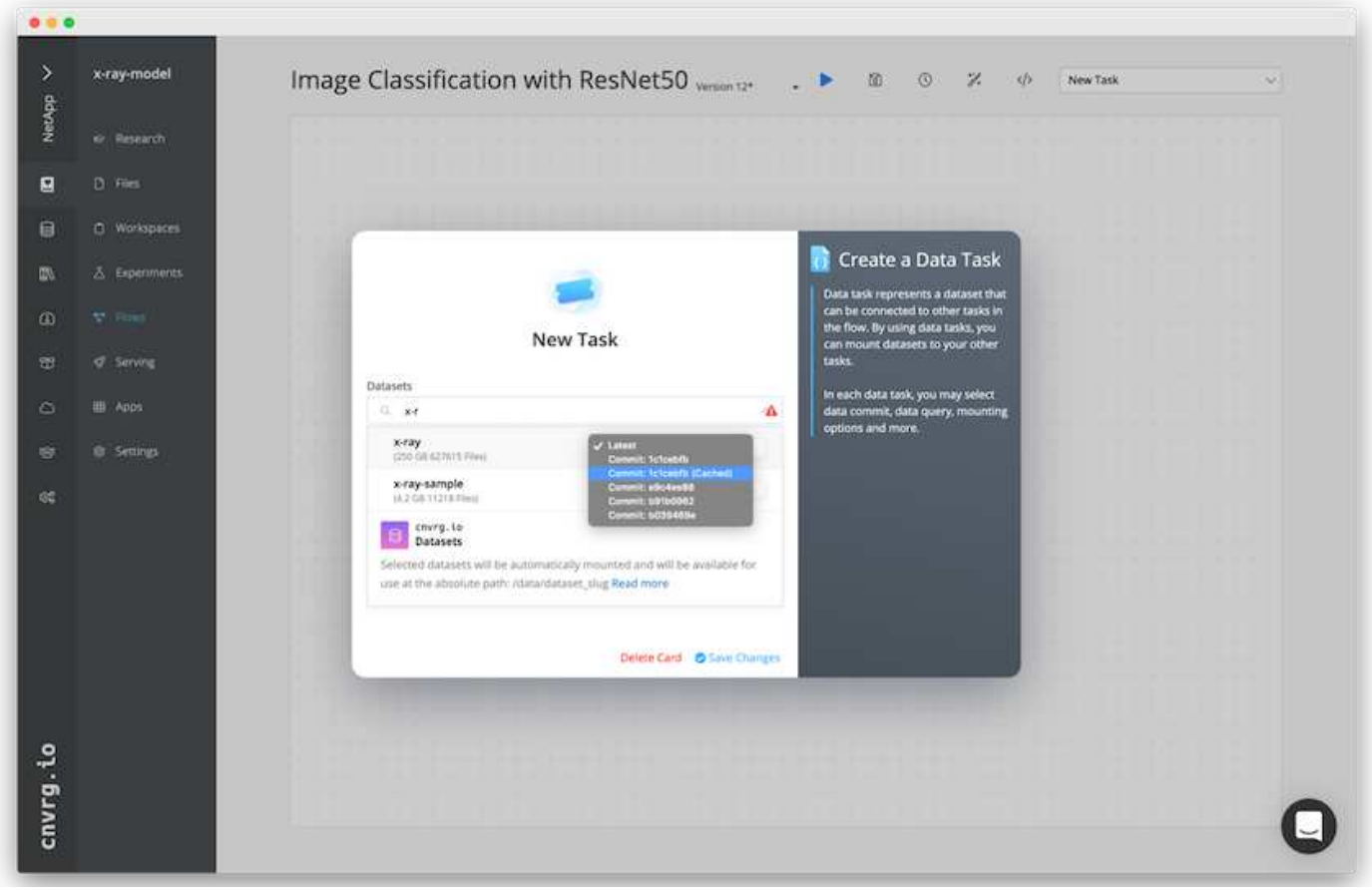
Build an ML Pipeline with Cached Data

cnvrg flows allows you to easily build production ML pipelines. Flows are flexible, can work for any kind of ML use case, and can be created through the GUI or code. Each component in a flow can run on a different compute resource with a different Docker image, which makes it possible to build hybrid cloud and optimized ML pipelines.



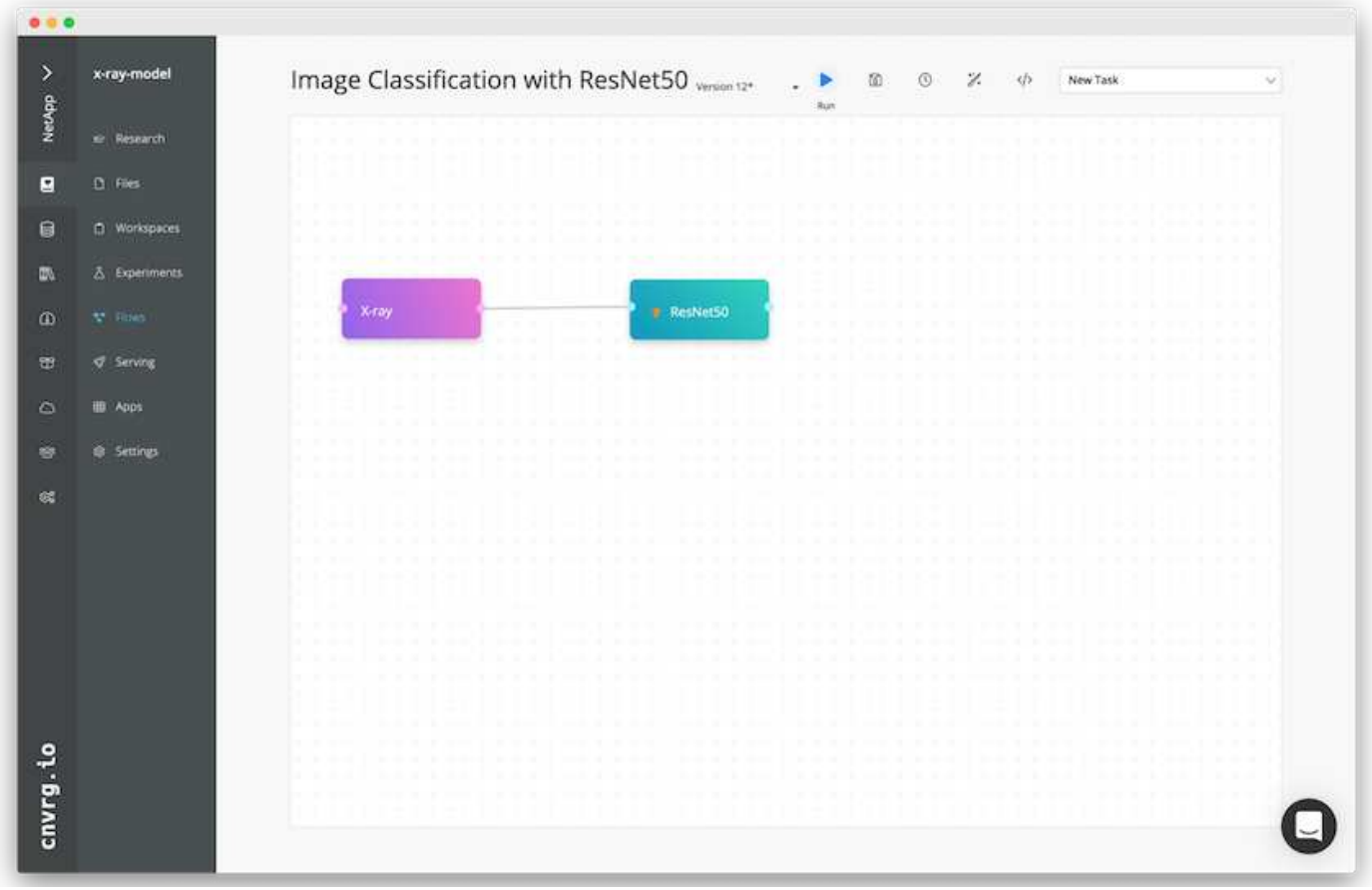
Building the Chest X-ray Flow: Setting Data

We added our dataset to a newly created flow. When adding the dataset, you can select the specific version (commit) and indicate whether you want the cached version. In this example, we selected the cached commit.



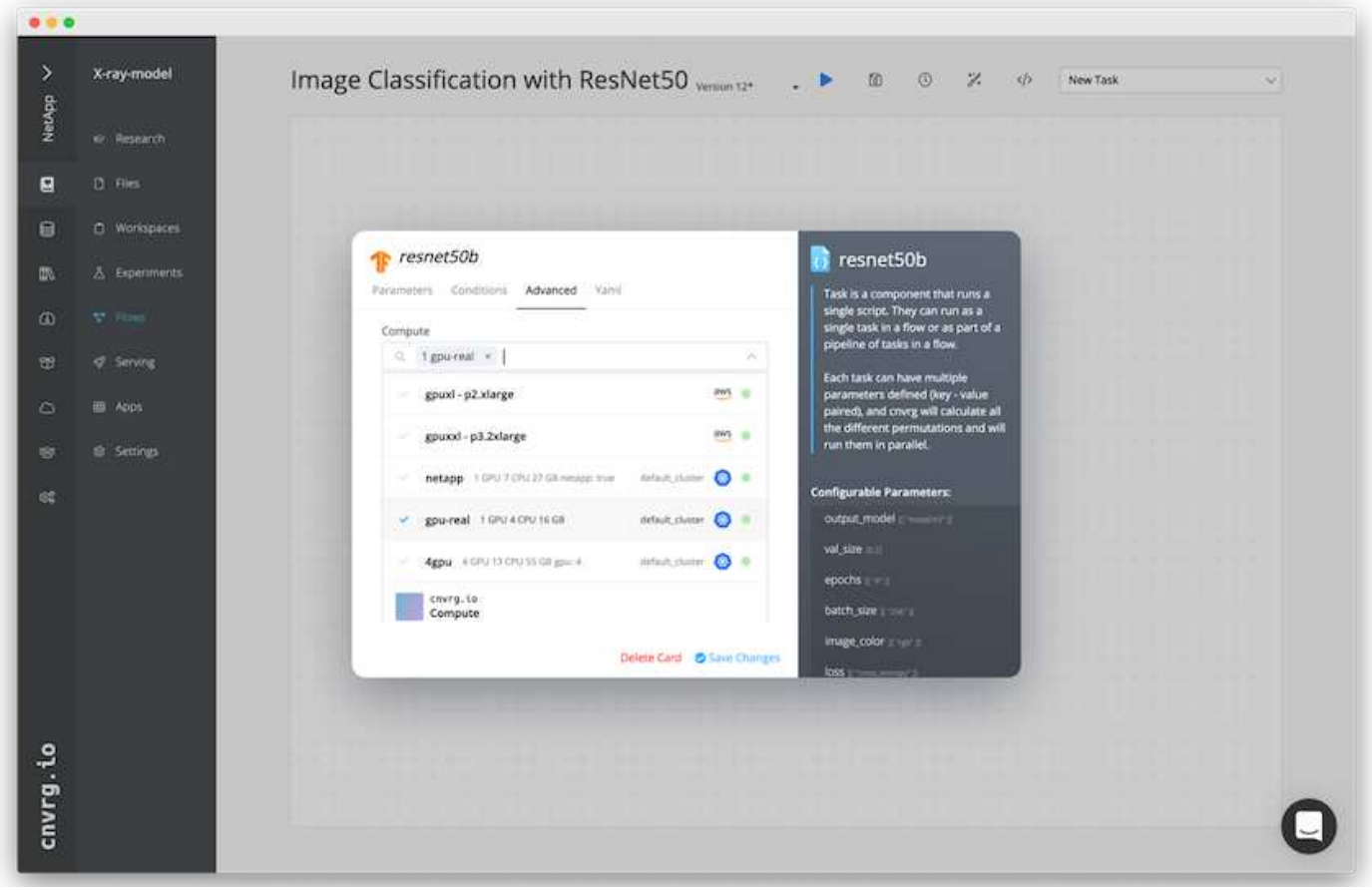
Building the Chest X-ray Flow: Setting Training Model: ResNet50

In the pipeline, you can add any kind of custom code you want. In cnvrg, there is also the AI library, a reusable ML components collection. In the AI library, there are algorithms, scripts, data sources, and other solutions that can be used in any ML or deep learning flow. In this example, we selected the prebuilt ResNet50 module. We used default parameters such as batch_size:128, epochs:10, and more. These parameters can be viewed in the AI Library docs. The following screenshot shows the new flow with the X-ray dataset connected to ResNet50.



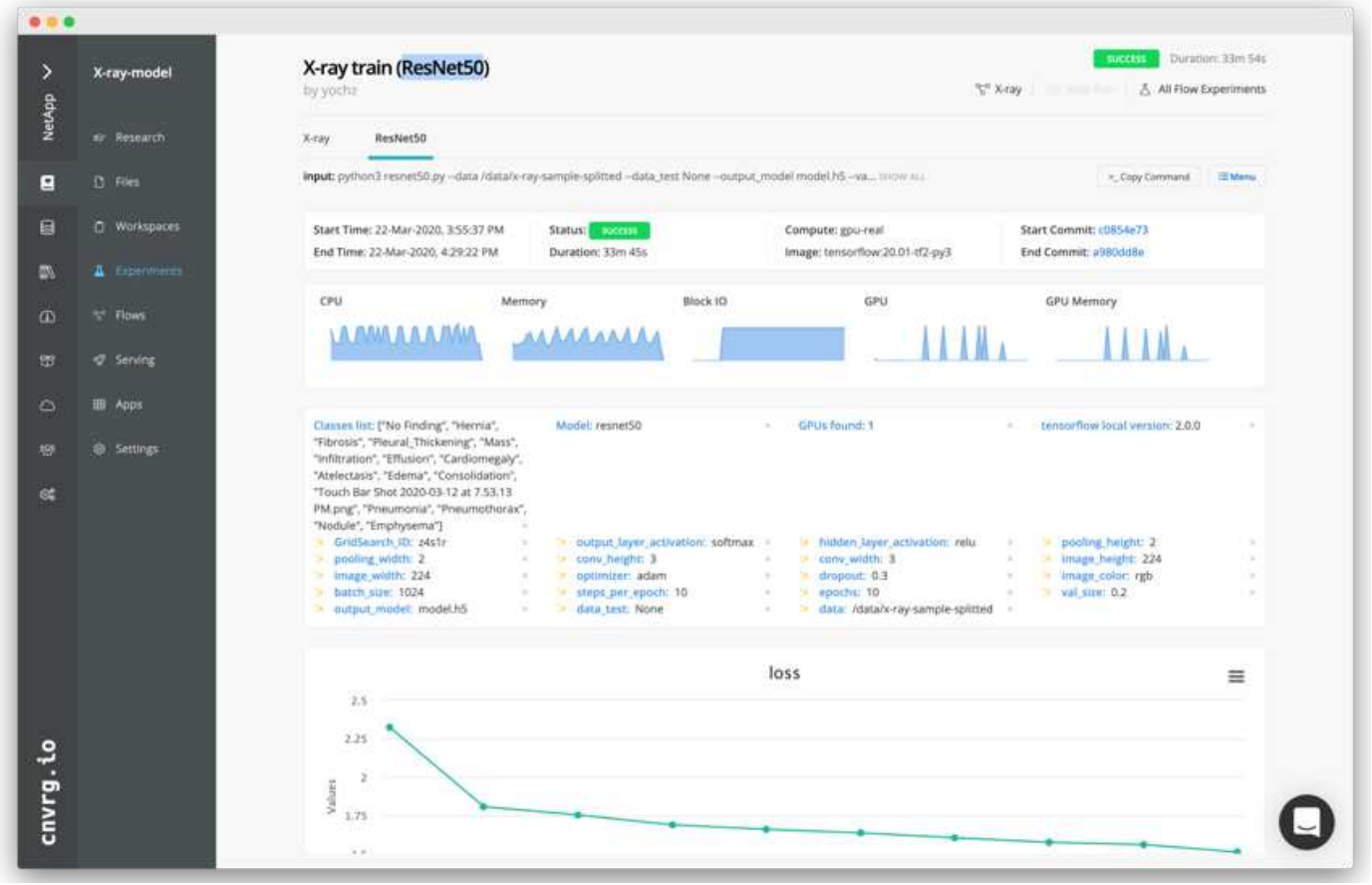
Define the Compute Resource for ResNet50

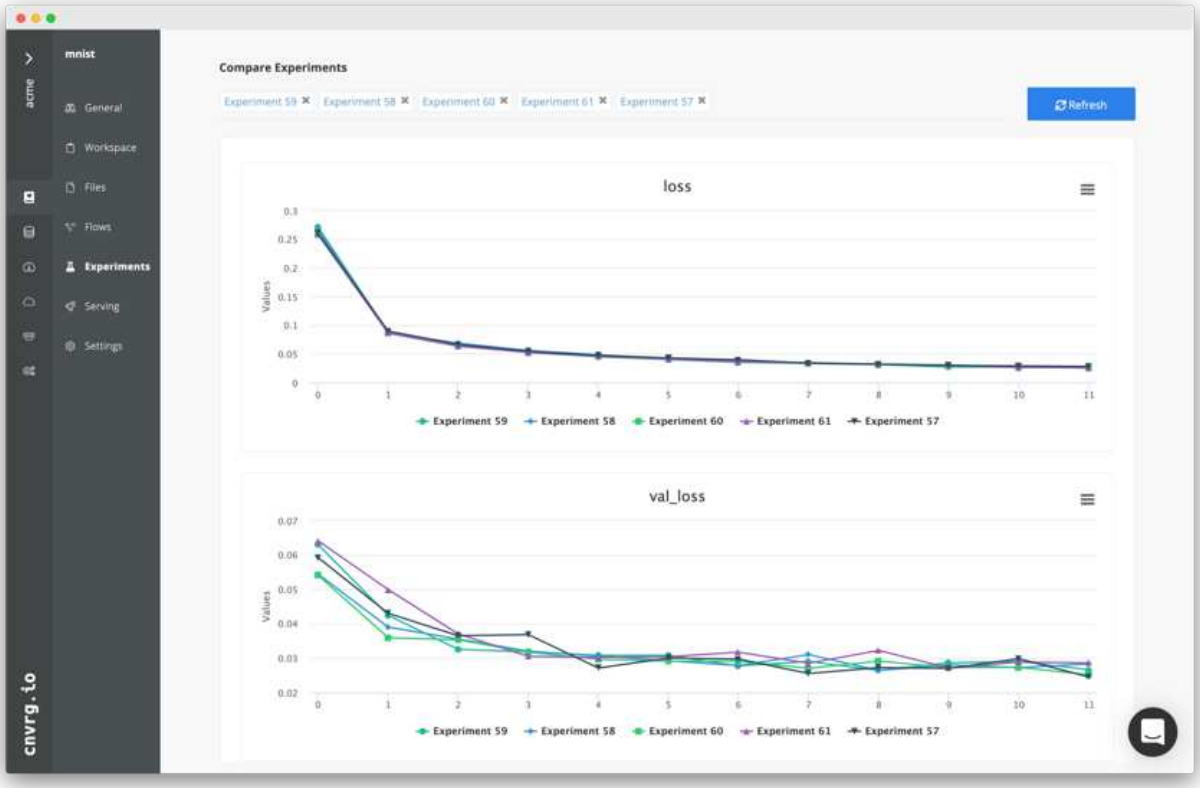
Each algorithm or component in cnvrg flows can run on a different compute instance, with a different Docker image. In our setup, we wanted to run the training algorithm on the NVIDIA DGX systems with the NetApp ONTAP AI architecture. In The following figure, we selected `gpu-real`, which is a compute template and specification for our on-premises cluster. We also created a queue of templates and selected multiple templates. In this way, if the `gpu-real` resource cannot be allocated (if, for example, other data scientists are using it), then you can enable automatic cloud-bursting by adding a cloud provider template. The following screenshot shows the use of gpu-real as a compute node for ResNet50.



Tracking and Monitoring Results

After a flow is executed, cnvrg triggers the tracking and monitoring engine. Each run of a flow is automatically documented and updated in real time. Hyperparameters, metrics, resource usage (GPU utilization, and more), code version, artifacts, logs, and so on are automatically available in the Experiments section, as shown in the following two screenshots.


NetApp
X-ray-model
Research
Files
Workspaces
Experiments
Flows
Serving
Apps
Settings
cnvrg.io
X-ray train (ResNet50)
by yochz
SUCCESS
Duration: 33m 54s
X-ray
All Flow Experiments
X-ray
ResNet50
input: python3 resnet50.py --data /data/x-ray-sample-splitted --data_test None --output_model model.h5 --va...
Copy Command
Menu
Start Time: 22-Mar-2020, 3:55:37 PM
End Time: 22-Mar-2020, 4:29:22 PM
Status: SUCCESS
Duration: 33m 45s
Compute: gpu-real
Image: tensorflow:20.01-tf2-py3
Start Commit: c0854e73
End Commit: a980dd8e
CPU
Memory
Block IO
GPU
GPU Memory
Classes list: ["No Finding", "Hernia", "Fibrosis", "Pleural_Thickening", "Mass", "Infiltration", "Effusion", "Cardiomegaly", "Atelectasis", "Edema", "Consolidation", "Touch Bar Shot 2020-03-12 at 7.53.13 PM.png", "Pneumonia", "Pneumothorax", "Nodule", "Emphysema"]
Model: resnet50
GPUs found: 1
tensorflow local version: 2.0.0
GridSearch_ID: z4s1r
pooling_width: 2
image_width: 224
batch_size: 1024
output_model: model.h5
output_layer_activation: softmax
conv_height: 3
optimizer: adam
steps_per_epoch: 10
data_test: None
hidden_layer_activation: relu
conv_width: 3
dropout: 0.3
epochs: 10
data: /data/x-ray-sample-splitted
pooling_height: 2
image_height: 224
image_color: rgb
val_size: 0.2
loss
Values
2.5
2.25
2
1.75


acme
mnist
General
Workspace
Files
Flows
Experiments
Serving
Settings
cnvrg.io
Compare Experiments
Experiment 59
Experiment 58
Experiment 60
Experiment 61
Experiment 57
Refresh
loss
Values
0.3
0.25
0.2
0.15
0.1
0.05
0
0 1 2 3 4 5 6 7 8 9 10 11
Experiment 59
Experiment 58
Experiment 60
Experiment 61
Experiment 57
val_loss
Values
0.07
0.06
0.05
0.04
0.03
0.02
0 1 2 3 4 5 6 7 8 9 10 11
Experiment 59
Experiment 58
Experiment 60
Experiment 61
Experiment 57

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